

IN-LIFE OPERATIONS

SYSTEM INSTALLATION AND CONFIGURATION ON AN HPE ALLETRA 4120 SERVER

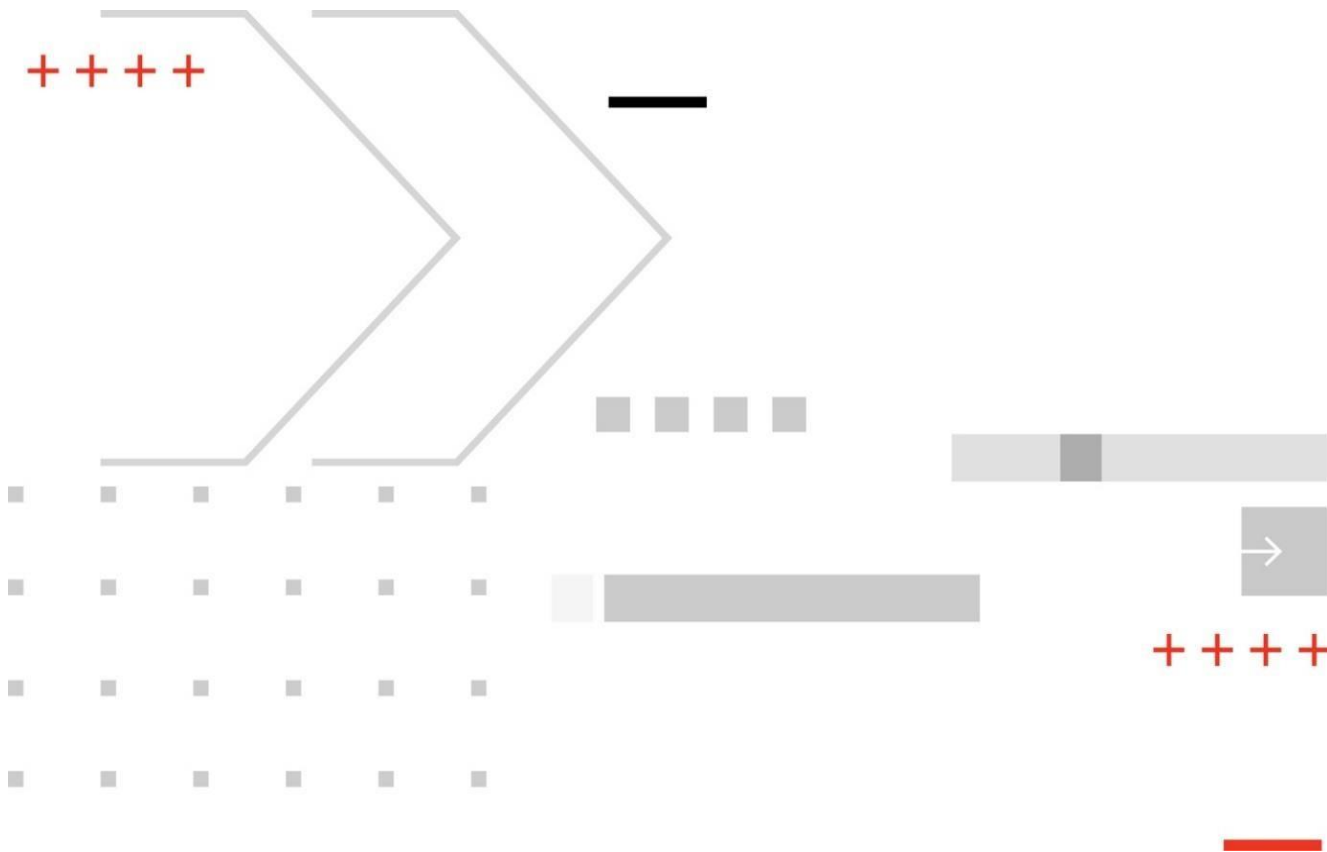
SYNOPSIS: This document describes the activities necessary for system configuration on an HPE Alletra 4120 server

REFERENCE: 100373 – Installation and service configuration manual

VERSION: V1.3

DATE: November 2025

STATUS: In Progress



IN-LIFE OPERATIONS
VEEAM BACKUP CONFIGURATION FOR IMMUTABLE REPOSITORY

1. INTRODUCTION

1.1 PURPOSE

This document describes the activities necessary to carry out the installation and configuration of the system on an HPE Alletra 4120 server, which will be used as a hardened repository in a backup solution based on Veeam Backup & Replication.

1.2 SCOPE

Ensure backup copies stored on the server for the protection of critical data, following configuration best practices and security guidelines recommended by both HPE and Veeam.

1.3 INTENDED AUDIENCE

This document is intended for use in services for Claro Colombia and Claro Cenam by the Product Support team (L1), Managed services team and L2 Support team.

The main activities carried out during this process were as follows:

1. Collection and validation of hardware and software requirements, verifying server compatibility with the operating system to be installed and with Veeam backup tools.
2. Creation of bootable installation media, using the official HPE utility to generate a USB with the Service Pack for ProLiant (SPP), facilitating the installation of the necessary operating system and firmware.
3. Installation of the Linux operating system (preferably Ubuntu Server LTS), following recommendations for secure backup servers.
4. Basic system configuration: hostname assignment, time zone, users and initial operating system updates.
5. Static network configuration and interface adjustment to ensure connectivity within the corporate network.
6. Firewall configuration and access rules, allowing only the necessary ports for communication between the backup server and the repository, ensuring a controlled and secure environment.
7. Preparation of local storage, including partition creation, volume mounting and permission adjustment for backup storage.
8. Repository hardening, applying security measures such as SSH root deactivation, key authentication implementation, and protection against direct backup deletion.
9. Repository registration in the Veeam console, performing the necessary configurations for the integration of the new Linux node as a secure backup destination.
10. Execution of a functional and communication test plan, to validate connectivity, storage and data transfer between solution components.
11. Documentation and backup of configurations performed, including diagrams, screenshots, configuration files and traceability of changes applied to the system.

This set of activities constitutes the basis for a clean and secure installation of the backup environment, with emphasis on the integrity, availability and protection of data managed through the HPE Alletra 4120 server and the Veeam solution.

2. ESSENTIAL SERVER CONCEPTS

A Below are the main essential concepts associated with the installation and configuration service of the HPE Alletra 4120 server in the context of a backup solution with Veeam Backup & Replication:

Concept	Description
HPE Alletra 4120	High-performance storage server designed for critical environments. Ideal for backup solutions and intensive workloads.
Service Pack for ProLiant (SPP)	Set of firmware, drivers and HPE tools for server installation, configuration and update. Used to create boot media.
Hardened Repository & Replication	Linux repository protected against tampering. Prevents deletion or modification of backups thanks to advanced security configurations.
Sistema Operativo Linux	Repository base. It is recommended to use LTS distributions (Ubuntu or Debian) for their stability and compatibility with Veeam.
SSH (Secure Shell)	Secure remote administration protocol. Allows connection to the server for configurations, monitoring and data transfer
Firewall	Network traffic control system. Allows configuring rules to allow or deny access, protecting the server from attacks or unauthorized access.
Veeam Communication Ports	Ports required by Veeam to interact with repositories (SSH: 22)
Access Control	Measures that restrict which users or services can interact with the server. Fundamental to ensure backup integrity.
Immutable Retention	Veeam functionality that protects backups against deletion for a specified period, even if system access is compromised.
Veeam Backup & Replication	Backup and data recovery software for virtual, physical and cloud environments. Allows managing and automating Backup policies from a central console.

3. HARDWARE AND SOFTWARE REQUIREMENTS

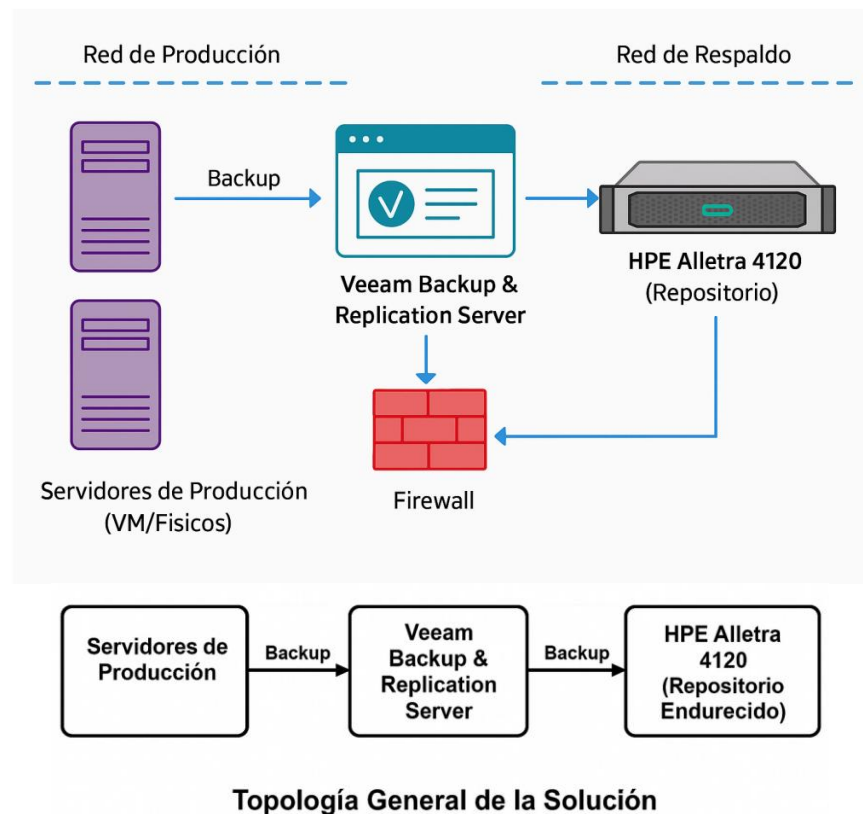
Below are the necessary requirements for service implementation, detailing both physical components and software tools used in each stage of the process:

Concept	Description
Servidor HPE Alletra 4120	High-performance storage server designed for critical environments. Ideal for backup solutions and intensive workloads.
RAM memory modules (minimum 16 GB)	Set of firmware, drivers and HPE tools for server installation, configuration and update. Used to create boot media.
Hard drives (preferably in RAID)	Linux repository protected against tampering. Prevents deletion or modification of backups thanks to advanced security configurations.
Network interface (minimum 1 Gbps)	Repository base. It is recommended to use LTS distributions (Ubuntu or Debian) for their stability and compatibility with Veeam.
Bootable USB with SPP or Linux ISO	Secure remote administration protocol. Allows connection to the server for configurations, monitoring and data transfer
Administration workstation	Network traffic control system. Allows configuring rules to allow or deny access, protecting the server from attacks or unauthorized access.
Ubuntu Server LTS (20.04 o Superior)	Ports required by Veeam to interact with repositories (SSH: 22)
Service Pack for ProLiant	Measures that restrict which users or services can interact with the server. Fundamental to ensure backup integrity.
Firewall (ufw or iptables)	Veeam functionality that protects backups against deletion for a specified period, even if system access is compromised.

4. SCHEMES AND TYPOLOGIES

General network topology:

The following topology illustrates the integration of the HPE Alletra 4120 server as a hardened repository within the backup infrastructure managed by Veeam Backup & Replication:



Production servers: virtual or physical machines containing the company's critical data.

Veeam Backup & Replication Server: server responsible for managing backup and recovery tasks. It communicates with production servers to obtain data and stores it in the repository.

HPE Alletra 4120 (Hardened Repository): server configured with advanced security measures to store backups immutably, protecting them against unauthorized modifications or deletions.

Network and Security Diagram

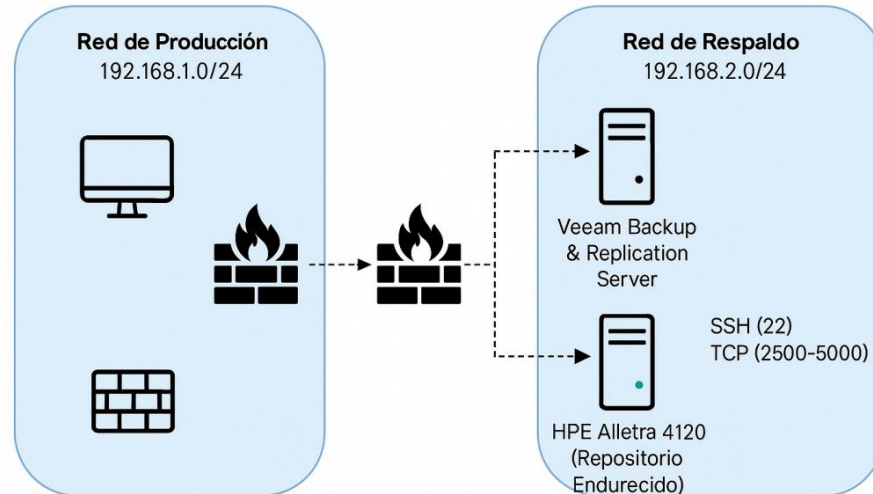
For risk mitigation in Backup operation, the use of Firewall is established with the following components in the network architecture:

Production network: Network segment where production servers reside.

Backup network: Network segment dedicated to backup operations, isolated from the production

network for greater security.

Firewall: Device or software that controls traffic between networks, allowing only the communications necessary for backup operations.



SSH traffic (port 22) is allowed from the Veeam server to the HPE Alletra 4120 for initial configuration and TCP port traffic 2500-5000 for backup data transfer.

All types of non-essential incoming traffic to the hardened repository are denied.

5. ESQUEMAS Y TIPOLOGIAS

5.1 Hardware Preparation

- Unpacking and component verification: ensure that all server components are present and in good condition.
- Rack mounting: install the server in the corresponding rack, following space and ventilation recommendations.
- Cable connection: connect power and network cables according to manufacturer specifications.

5.2 Firmware Update

- Download of Service Pack for ProLiant (SPP): obtain the latest version of SPP from the official HPE site, as shown in the following figure:

Service pack for HPE ProLiant Gen12 version 2025.03.00.00

Download Link: [HPE ProLiant Gen12](#)

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Descripción general del SPP

Tipo	Fecha de lanzamiento	Número de pieza	Paquetes de parches
Base SPP	2 de abril de 2025	P81144-001	
Versión SUM	Tamaño		
12.0.0	1,58 GB		
Actualización de la versión SPP			
Gen12 2025.01.00.00, 2024.11.00.00			
Crear una descarga personalizada de SPP			
Descargar Base SPP			

Notas importantes

El paquete de servicios para HPE ProLiant Gen12 versión 2025.03.00.00 incluye soporte para las siguientes nuevas plataformas de servidor Gen12 basadas en Intel basadas en HPE Integrated Lights-Out 7:

- Servidor HPE ProLiant Compute DL320 Gen12
- Servidor HPE ProLiant Compute DL340 Gen12
- HPE ProLiant Compute DL360 Gen12
- HPE ProLiant Compute DL380 Gen12
- HPE ProLiant Compute DL380a Gen12 (U72)
- Servidor HPE ProLiant ML350 Gen12

Leer más

Resumen del lanzamiento

El paquete de servicios para la versión de producción 2025.03.00.00 de HPE ProLiant Gen12 contiene componentes de firmware, software y controladores compatibles con las plataformas de servidor HPE ProLiant Gen12 (basadas en HPE Integrated Lights-Out 7 y 6) y opciones.

Esta versión del paquete de servicio Gen12 incluye Smart Update Manager (SUM) 12.0.0 junto con HPE iLO7 versión 1.12.00 y HPE iLO6 versión 1.67.

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Software

Drivers, Patches & Firmware

SPP Gen11

SPP Gen10

SPP All Gen

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Service Pack for ProLiant 2025.03 Gen12

Details

Product Number:
SPPGen12.2025.03.00.00_Base

Category:
Software

Product Family:
Service Pack for ProLiant

Software Type:
Software

Version:
2025.03.00.00
Jun 24, 2025

Save yourself the time of downloading SPPs and creating custom scripts. HPE GreenLake for Compute Ops Management does the work for you. This subscription-based SaaS tool allows you to update and manage all your servers from a central, cloud-based console. Update the firmware of tens to thousands of servers at a time with a couple clicks. Learn more at <https://hes.com/info/COM>.

Software (3)

P81144_001_gen12spp-2025.03.00.00-Gen12SPP2025030000.2025_0328.39.iso (1.58 GB)

SHA256 Checksum: b3d5102c68fa01cd5c931576e6aa227201c009ca85b08b777... (Copy)

sha2sum_gen12spp-2025.03.00.00-Gen12SPP2025030000.2025_0328.39.iso.sha2sum (102 Bytes)

SHA256 Checksum: eec7452b345c1aae0113421a262f5ef5934f0d0db279fee4b... (Copy)

md5sum_gen12spp-2025.03.00.00-Gen12SPP2025030000.2025_0328.39.iso.md5sum (70 Bytes)

SHA256 Checksum: 34f18fb8577c6769b0a915d86fc9594a799259388361f3854a... (Copy)

Your download speed may be affected by your internet service provider (ISP).

Download

curl Copy

- Boot media creation: use a tool like Rufus to create a bootable USB with the SPP. For bootable USB creation.
- Proceed with the creation of a bootable USB following the steps below:



5.3 Veeam Hardened Repository Installation

The procedure is to download the ISO from the Veeam Hardened Repository: obtain the official ISO image from the Veeam portal.

Creation of installation media preparing a bootable USB with the ISO image, it is required to start the SPP tool selecting automatic mode. The steps to follow in the download will be current inventory review > Process review > Deployment.

When the installation process is complete, the server is restarted to start with the configurations established by Veeam.

The operating system is installed by booting the server from USB and following the installation wizard steps, configuring parameters such as time zone, network and disk partitioning.

5.4 Security and Access Configuration

The creation of users with limited privileges for repository management is established.

Retention policies are configured: defining backup retention policies and enabling immutability to protect data against unauthorized modifications or deletions.

Network firewall settings, configuring firewall rules and network access to ensure communication between server and Veeam components.

6. SECURITY ASPECTS

Ports Required by Veeam

To ensure proper communication between Veeam components and the hardened repository, it is necessary to allow traffic through the following ports:

Origen	Destino	Protocolo	Puerto(s)	Description
Veeam Backup Console	Backup server	TCP	9392, 9420	Communication between the console and the backup server.
Backup server	Hardened repository (Linux)	TCP	2500-3300	Data transmission channels for backup and restore jobs
Backup server	Hardened repository (Linux)	TCP	22	SSH connection for management and data transfer.
Backup server	Hardened repository (Linux)	TCP	6160	Communication of the backup agent with the backup server.

6.1 Firewall Configuration on Hardened Repository (Linux)

If using a Linux distribution like Ubuntu for the hardened repository, you can configure the firewall using ufw (Uncomplicated Firewall)

```
sudo ufw allow 22/tcp  
Sudo ufw allow 2500:3300/tcp  
Sudo ufw allow 6160/tcp  
Sudo ufw enable
```

These commands allow the necessary connections for Veeam to interact with the repository.

6.2 Firewall Configuration on Backup Server (Windows)

On the backup server running Veeam Backup & Replication on Windows, you must ensure that the firewall allows outbound connections to the ports mentioned above.

You can use the following PowerShell command to allow traffic to port 2500, for example:

New-NetFirewallRule -DisplayName “Veeam Data ports” -Direction Outbound -Protocol TCP -LocalPort:

Repeat this process for other necessary ports.

6.3 Connectivity Verification

After configuring the firewall rules, verify connectivity between the backup server and the hardened repository:

- From the backup server (Windows)
Test-NetConnection – ComputerName [IP-del-repositorio] –Port 22
- From the hardened repository (Linux)
nx -zv [IP_del_servidor_de_respaldo] 9392

These commands will help confirm that the necessary ports are open and that communication is possible.

6.4 Consideraciones Adicionales de Seguridad

- SSH Access: limit SSH access to the hardened repository only from the backup server and other trusted sources.
- Updates: keep both the backup server and the hardened repository updated with the latest security patches.
- Monitoring: Implement monitoring tools to detect unauthorized access or suspicious activity

Version control

Version	Fecha	Author	Description
1.0	07/10/2025	Pablo César G.	Solution general topology
1.1	14/10/2025	Pablo César G.	Veeam hardened repository installation
1.2	21/10/2025	Pablo César G.	Firewall Configuration on Repository

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